

Normalization

This document presents the normalization process for the Hospital Management System (HMS) database. The database is normalized from Un-normalized Form (UNF) to Boyce-Codd Normal Form (BCNF) in order to reduce redundancy, eliminate anomalies, and improve data integrity.

Un-Normalized Form (UNF)

Before normalization, all hospital information exists in a single relation.

HOSPITAL(
PatientID, FullName, Gender, DateOfBirth, Phone, Address,
DoctorID, DoctorName, Specialization,
DepartmentID, DepartmentName, Location,
AppointmentID, AppointmentDate, AppointmentStatus,
RecordID, RecordDate, Diagnosis, Treatment,
PrescriptionID, Medication, Dosage, Instructions,
Bill_ID, TotalAmount, BillDate,
PaymentID, AmountPaid, PaymentDate, PaymentMethod,
CaseID, SeverityLevel, ArrivalTime, TreatmentGiven,
QueueID, QueueNumber
)

Example UNF

PatientID	FullName	DoctorName	AppointmentDate	Diagnosis	BillAmount
P001	Hana Mekonnen	Dr. Bekele	2026-05-10	Hypertension	1500.00
P002	Dawit Tadesse	Dr. Alemu	2026-05-12	Flu	500.00
P003	Meron Getachew	Dr. Sifan	2026-05-13	Diabetes	2000.00

Problems in UNF

- ✓ **Redundancy:** Patient and doctor names repeat across rows.
- ✓ **Repeating groups:** Multiple appointments or bills tied to one patient.
- ✓ **Update anomalies:** Changing doctor info requires updating every row.
- ✓ **Insertion anomalies:** Cannot add a new doctor without a patient.
- ✓ **Deletion anomalies:** Removing a bill may delete patient data.

Step 1: First Normal Form (1NF)

Rule

- ✧ All attributes contain atomic values
- ✧ There are no repeating groups
- ✧ Each record is unique

Functional Dependencies:

PatientID → FullName, Gender, DateOfBirth, Phone, Address

DoctorID → DoctorName, Specialization, DepartmentID

DepartmentID → DepartmentName, Location

AppointmentID → AppointmentDate, AppointmentStatus, PatientID, DoctorID

RecordID → RecordDate, Diagnosis, Treatment, PatientID

PrescriptionID → Medication, Dosage, Instructions, RecordID

Bill_ID → TotalAmount, BillDate, PatientID

PaymentID → AmountPaid, PaymentDate, PaymentMethod, Bill_ID

CaseID → SeverityLevel, ArrivalTime, TreatmentGiven, PatientID

QueueID → QueueNumber, AppointmentID

Relations after 1NF

Relation	Attributes
Patient	PatientID(PK), FullName, Gender, DateOfBirth, Phone, Address
Doctor	DoctorID(PK), DoctorName, Specialization, DepartmentID(FK)
Department	DepartmentID(PK), DepartmentName, Location
Appointment	AppointmentID(PK), AppointmentDate, AppointmentStatus, PatientID(FK), DoctorID(FK)
Medical_Record	RecordID(PK), RecordDate, Diagnosis, Treatment, PatientID(FK)
Prescription	PrescriptionID(PK), Medication, Dosage, Instructions, RecordID(FK)
Billing	Bill_ID(PK), TotalAmount, BillDate, PatientID(FK)
Payment	PaymentID(PK), AmountPaid, PaymentDate, PaymentMethod, Bill_ID(FK)
Emergency_Case	CaseID(PK), SeverityLevel, ArrivalTime, TreatmentGiven, PatientID(FK)

Queue	QueueID(PK), QueueNumber, AppointmentID(FK)
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Step 2: Second Normal Form (2NF)

Rule

- ✧ Relation must already be in 1NF
- ✧ All non-key attributes fully depend on the entire primary key
- ✧ There are no partial dependencies

2NF Analysis

All relations in the Hospital Management System use single primary keys. Since there are no composite primary keys, partial dependency does not exist. Therefore, all relations satisfy Second Normal Form (2NF).

Schema after 2NF

Relation	Primary Key	Foreign keys
Patient	PatientID	-----
Doctor	DoctorID	DepartmentID
Department	DepartmentID	-----
Appointment	AppointmentID	PatientID, DoctorID
Medical_Record	RecordID	PatientID
Prescription	PrescriptionID	RecordID
Billing	Bill_ID	PatientID
Payment	PaymentID	Bill_ID
Emergency_Case	CaseID	PatientID
Queue	QueueID	AppointmentID

NB: No partial dependencies exist. Therefore 2NF is satisfied.

Step 3: Third Normal Form (3NF)

Rule

- ✧ Relation must already be in 2NF
- ✧ There are no transitive dependencies
- ✧ Non-key attributes depend only on the primary key

Transitive Dependency Analysis

DoctorID → DepartmentID

DepartmentID → DepartmentName, Location

Thus: DoctorID → DepartmentName, Location

To eliminate this dependency, the Department relation is separated from the Doctor relation.

Relations in 3NF

Relation	Attributes	Dependency Check	Result
Patient	PatientID, FullName, Gender, DateOfBirth, Phone, Address	All attributes depend directly on PatientID	OK
Doctor	DoctorID, DoctorName, Specialization, DepartmentID	Department details stored separately in Department table	OK
Department	DepartmentID, DepartmentName, Location	All attributes depend directly on DepartmentID	OK
Appointment	AppointmentID, AppointmentDate, AppointmentStatus, PatientID, DoctorID	All attributes depend directly on AppointmentID	OK
Medical_Record	RecordID, RecordDate, Diagnosis, Treatment, PatientID	All attributes depend directly on RecordID	OK
Prescription	PrescriptionID, Medication, Dosage, Instructions, RecordID	All attributes depend directly on PrescriptionID	OK
Billing	Bill_ID, TotalAmount, BillDate, PatientID	All attributes depend directly on Bill_ID	OK
Payment	PaymentID, AmountPaid, PaymentDate, PaymentMethod, Bill_ID	All attributes depend directly on PaymentID	OK
Emergency_Case	CaseID, SeverityLevel, ArrivalTime, TreatmentGiven, PatientID	All attributes depend directly on CaseID	OK
Queue	QueueID, QueueNumber, AppointmentID	All attributes depend directly on QueueID	OK

Step 4: Boyce-Codd Normal Form (BCNF)

A relation is said to be in Boyce-Codd Normal Form (BCNF) if for every non-trivial functional dependency $X \rightarrow Y$, X must be a candidate key.

BCNF helps improve:

- ✓ data consistency
- ✓ data integrity
- ✓ redundancy reduction

BCNF Verification Table

<i>Relation</i>	<i>Determinant</i>	<i>Candidate Key</i>	<i>BCNF Status</i>
Patient	PatientID	PatientID	Satisfies BCNF
Doctor	DoctorID	DoctorID	Satisfies BCNF
Department	DepartmentID	DepartmentID	Satisfies BCNF
Appointment	AppointmentID	AppointmentID	Satisfies BCNF
Medical_Record	RecordID	RecordID	Satisfies BCNF
Prescription	PrescriptionID	PrescriptionID	Satisfies BCNF
Billing	Bill_ID	Bill_ID	Satisfies BCNF
Payment	PaymentID	PaymentID	Satisfies BCNF
Emergency_Case	CaseID	CaseID	Satisfies BCNF
Queue	QueueID	QueueID	Satisfies BCNF

All relations in the Hospital Management System satisfy Boyce-Codd Normal Form (BCNF) because every determinant is a candidate key.

Final Normalized Schema**PATIENT**

<i>Attributes</i>	<i>Type</i>	<i>Key</i>	<i>Description</i>
PatientID	INT	PK	Unique patient identifier
FullName	VARCHAR(100)		Patient's full name
Gender	VARCHAR(10)		Gender of patient
DateOfBirth	DATE		Birth date
Phone	VARCHAR(20)		Contact number
Address	VARCHAR(200)		Residential address

DOCTOR

<i>Attributes</i>	<i>Type</i>	<i>Key</i>	<i>Description</i>
DoctorID	INT	PK	Unique doctor identifier
DoctorName	VARCHAR(100)		Doctor's full name
Specialization	VARCHAR(50)		Medical specialty

DepartmentID	INT	FK	References DEPARTMENT
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DEPARTMENT

<i>Attributes</i>	<i>Type</i>	<i>Key</i>	<i>Description</i>
DepartmentID	INT	PK	Unique department identifier
DepartmentName	VARCHAR(100)		Department name
Location	VARCHAR(100)		Department location

APPOINTMENT

<i>Attributes</i>	<i>Type</i>	<i>Key</i>	<i>Description</i>
AppointmentID	INT	PK	Unique appointment identifier
AppointmentDate	DATE		Date of appointment
AppointmentStatus	VARCHAR(20)		Scheduled / Completed / Cancelled
PatientID	INT	FK	References PATIENT
DoctorID	INT	FK	References DOCTOR

MEDICAL_RECORD

<i>Attributes</i>	<i>Type</i>	<i>Key</i>	<i>Description</i>
RecordID	INT	PK	Unique medical record identifier
RecordDate	DATE		Date of record entry
Diagnosis	VARCHAR(200)		Patient diagnosis
Treatment	VARCHAR(200)		Treatment details
PatientID	INT	FK	References PATIENT

PRESCRIPTION

<i>Attributes</i>	<i>Type</i>	<i>Key</i>	<i>Description</i>
PrescriptionID	INT	PK	Unique prescription identifier
Medication	VARCHAR(100)		Prescribed medicine
Dosage	VARCHAR(50)		Dosage instructions
Instructions	VARCHAR(200)		Additional instructions
RecordID	INT	FK	References MEDICAL_RECORD

BILLING

<i>Attributes</i>	<i>Type</i>	<i>Key</i>	<i>Description</i>
Bill_ID	INT	PK	Unique bill identifier
TotalAmount	DECIMAL(12,2)		Total bill amount
BillDate	DATE		Date of bill
PatientID	INT	FK	References PATIENT

PAYMENT

<i>Attributes</i>	<i>Type</i>	<i>Key</i>	<i>Description</i>
PaymentID	INT	PK	Unique payment identifier
AmountPaid	DECIMAL(12,2)		Amount paid
PaymentDate	DATE		Date of payment
PaymentMethod	VARCHAR(50)		Cash / Card / Insurance
Bill_ID	INT	FK	References BILLING

EMERGENCY_CASE

<i>Attributes</i>	<i>Type</i>	<i>Key</i>	<i>Description</i>
CaseID	INT	PK	Unique emergency case identifier
SeverityLevel	VARCHAR(20)		Severity of case
ArrivalTime	DATETIME		Time of arrival
TreatmentGiven	VARCHAR(200)		Treatment provided
PatientID	INT	FK	References PATIENT

QUEUE

<i>Attributes</i>	<i>Type</i>	<i>Key</i>	<i>Description</i>
QueueID	INT	PK	Unique queue identifier
QueueNumber	INT		Queue position
AppointmentID	INT	FK	References APPOINTMENT

Conclusion

The Hospital Management System database was successfully normalized from Un-normalized Form (UNF) to Boyce-Codd Normal Form (BCNF). During the normalization process, repeating groups, partial dependencies, and transitive dependencies were removed to improve data consistency and reduce redundancy.

The final relations are properly structured and support efficient data organization for hospital operations such as patient management, appointments, billing, prescriptions, and emergency cases.